A

Industrial Training/Internship Report

On

VOCATIONAL TRAINING IN MACHINE LEARNING & COMPUTER VISION

REPORT

Submitted to

CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI

in partial fulfilment of requirement for the award of the degree

of

Bachelor of Technology

in

Computer Science and Engineering

by

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BHILAI INSTITUTE OF TECHNOLOGY, DURG (CG)

(Seth Balkrishan Memorial) Estd. 1986

ALL UG & MBA COURSES | NAAC 'A' GRADE | ISO 9001:2015 | ISO 14001:2015 | NIRF

Certificate by Company / Industry

Certificate of Completion issued by BS Digital Technology

BS DIGITAL TECHNOLOGY

CERTIFICATE OF COMPLETION

This is to certify that Shri / Ku. Harsh Patel
Reg. No.: BIT/2026/82
6th Sem., Student of B.Tech. of Bhilai Institute of Technology College / Institute Durg

Has undergone project based training from 08/06/2026 to 08/07/2026

Project: HemoVision — Contactless Vital Signs Estimation via rPPG

Technology: Python, PyTorch, OpenCV, Deep Learning, ML

His / Her performance during the training period has been: Good

Mr. Raahul Kharatkar **HR Manager** **BS Digital Technology**	**Mr. Prashant Kumar Tamarakar** **Managing Director** **BS Digital Technology**

Figure: Company/Industry Certificate of Completion

<u>Declaration by Student</u>

I, Harsh Patel, a student of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence), 6th Semester, hereby declare that the Industrial Training / Vocational Training report entitled "HemoVision – Contactless Vital-Sign Estimation and Biomarker Analysis via Remote Photoplethysmography (rPPG)" is based on the work carried out by me during my project-based training at BS Digital Technology from 08 June 2026 to 08 July 2026.

I declare that the work presented in this report reflects my learning, implementation, analysis, and project activities during the training period. The report has been prepared for academic submission and has not been submitted elsewhere for the award of any other degree, diploma, or certificate.

Wherever information, concepts, tools, documentation, or external references have been used, appropriate acknowledgement and references have been provided. I have made every effort to present the project work accurately and in accordance with the prescribed university report format.

Signature of Student: ______________________________

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Roll No.: 300111323021
Enrolment No.: CD8812

Acknowledgement

I have great pleasure in the submission of this project report entitled "HemoVision – Contactless Vital-Sign Estimation and Biomarker Analysis via Remote Photoplethysmography (rPPG)" in partial fulfilment of Vocational Training. While submitting this project report, I take this opportunity to thank those directly or indirectly related to project work. I would like to thank my supervisor Mr Kailash Sinha and my vocational training incharge Dr Sumit Sar who has provided the opportunity and organizing project for me. Without his active co-operation and guidance, it would have become very difficult to complete the task in time. I would like to express sincere thanks and gratitude to Dr. Anup Mishra, Principal of the Institution, Dr. (Mrs.) Sunita Soni, Head of the Department Computer Science & Engineering for their encouragement and cordial support.

Acknowledgement is due to our parents, family members, friends and all those persons who have helped us directly or indirectly in the successful completion of the project work.

Name of the Student: Harsh Patel
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Abstract

HemoVision is a project-based computer vision and deep learning solution developed during vocational training at BS Digital Technology. The project focuses on contactless physiological vital-sign estimation and biomarker analysis from standard RGB facial video using Remote Photoplethysmography (rPPG). The solution combines facial landmark tracking, multi-region of interest (ROI) extraction, signal AC/DC disentanglement, temporal deep neural networks, and interactive clinical visualization into an end-to-end workflow.

The project uses the clinical MCD-rPPG dataset comprising 3,427 video clips across 598 unique subjects captured across three consumer camera types (FullHDwebcam, USBVideo, IriunWebcam) and two conditions (resting and post-exercise). The data preparation process includes tracking 468 3D landmarks using MediaPipe Face Mesh, segmenting 8 anatomically independent facial skin ROIs, extracting 24-channel RGB time-series, isolating AC pulsatile signals via z-score normalization, retaining raw DC luminance representations, and applying 2nd-order Butterworth bandpass filtering (0.65–3.25 Hz). Multiple temporal models were explored, and a lightweight Depthwise-Separable Bounded Temporal Convolutional Network (HemoVisionBoundedTCN) was selected as the final model for pulse waveform reconstruction and heart-rate estimation.

The trained solution achieved a Mean Absolute Error (MAE) of 9.19 BPM on a strictly subject-disjoint held-out test set of 91 subjects, with high-quality clips reaching 1.33 BPM MAE, outperforming classical signal processing baselines (POS MAE 19.17 BPM; CHROM MAE 20.28 BPM). Furthermore, an in-depth Biomarker Feasibility Audit demonstrated that RGB cameras cannot reliably reconstruct blood chemistry without dual-wavelength NIR hardware. An interactive Streamlit and OpenCV dashboard was developed to present real-time facial ROI tracking, BVP waveforms, and frequency spectral peaks. The project provided practical exposure to the complete computer vision and deep learning lifecycle.

Keywords: Remote Photoplethysmography (rPPG), Deep Learning, Temporal Convolutional Network (TCN), Computer Vision, MediaPipe Face Mesh, Vital Signs, Heart Rate Estimation, Biomarker Feasibility, PyTorch, OpenCV, Streamlit.

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Abbreviations and Nomenclature

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
rPPG	Remote Photoplethysmography
PPG	Photoplethysmography
BVP	Blood Volume Pulse
ECG	Electrocardiogram
ROI	Region of Interest
TCN	Temporal Convolutional Network
POS	Plane-Orthogonal-to-Skin rPPG algorithm
CHROM	Chrominance-based rPPG method
FFT	Fast Fourier Transform
PSD	Power Spectral Density
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
BPM	Beats Per Minute
SpO2	Peripheral Capillary Oxygen Saturation
BP	Blood Pressure (Systolic / Diastolic)
AMP	Automatic Mixed Precision
CCC	Lin's Concordance Correlation Coefficient
Streamlit	Python-based interactive web deployment framework
MediaPipe	Cross-platform framework for 468 3D Face Mesh perception

1. Introduction

Industrial and vocational training provides an important bridge between academic learning and practical application. For a Computer Science and Engineering student specializing in Artificial Intelligence, exposure to real biological datasets, computer vision pipelines, deep temporal learning models, and interactive clinical visualization tools is particularly valuable. The present training was completed at BS Digital Technology through a project-based program from 08 June 2026 to 08 July 2026.

The central project undertaken during the training was HemoVision, a contactless vital-sign estimation and biomarker analytics solution. The project was designed around a practical healthcare requirement: conventional vital-sign monitoring relies on intrusive contact sensors (pulse oximeters, blood pressure cuffs, ECG electrodes) and invasive blood sampling. Remote Photoplethysmography (rPPG) provides a non-invasive alternative by measuring sub-visual skin color fluctuations caused by cardiovascular blood pulses from standard RGB video cameras.

HemoVision follows an end-to-end workflow. Facial video is captured via consumer webcams and processed with MediaPipe Face Mesh to extract 8 anatomically stable facial Regions of Interest (ROIs). Python and PyTorch are then used for signal preprocessing, AC/DC disentanglement, Butterworth bandpass filtering, and temporal deep learning experimentation. The final pulse wave and heart rate predictions are presented through an interactive Streamlit clinical dashboard.

The project uses the clinical MCD-rPPG dataset covering 598 unique subjects across 3,427 video clips. Variables such as 468 facial landmark coordinates, 8 skin ROIs, camera sensor characteristics (FullHDwebcam, USBVideo, IriunWebcam), subject physical state (resting and post-exercise), and synchronized contact ground truth (ECG pulse, SpO2, blood pressure, respiratory rate, hemoglobin) help transform raw video streams into a structured analytical framework.

1.1 Training Profile and Project Details

Item	Details
Student	Harsh Patel
Program	B.Tech – Computer Science and Engineering (Artificial Intelligence)
Semester	6th Semester
Roll No.	300111323021
Enrolment No.	CD8812
Training Organization	BS Digital Technology
Project	HemoVision — Contactless Vital-Sign Estimation via rPPG
Training Type	Project-based vocational training
Training Period	08 June 2026 – 08 July 2026
Duration	30 days
Primary Technologies	Python, PyTorch, OpenCV, MediaPipe, Deep Learning
Deployment / UI Layer	Streamlit, SciPy, Matplotlib, Scikit-Learn

1.2 Project Overview

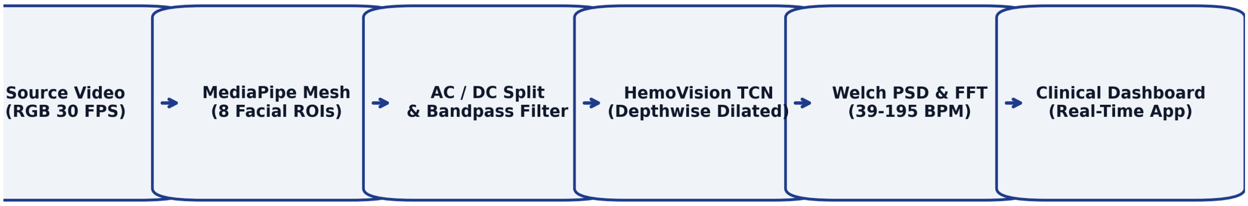

Figure 2: HemoVision End-to-End Workflow

The workflow starts with ambient facial video captures and ends with clinical-grade physiological forecasts. The modular separation between facial tracking, signal preprocessing, temporal neural modeling, and visualization makes HemoVision robust, easy to interpret, maintain, and extend into real-time telemedicine applications.

2. Formal Training Provided

The training was project-oriented and focused on connecting computer vision, biological signal processing, and deep learning concepts into a practical contactless health monitoring system. Rather than treating individual technologies as isolated topics, the learning process connected them through a complete workflow: video decoding, facial landmark mesh tracking, multi-ROI spatial signal extraction, digital filtering, PyTorch neural model development, and interactive web-based deployment.

2.1 Python and Computer Vision / Signal Processing

Python was used as the main programming environment for computer vision and signal processing. The training strengthened practical proficiency in libraries such as OpenCV, MediaPipe, NumPy, and SciPy for video decoding, spatial pixel pooling, and 1D digital signal analysis:

- Decoding raw video frames and handling camera stream buffers using OpenCV.
- Extracting 468 3D facial landmarks and dynamic region masking using MediaPipe Face Mesh.
- Implementing 8 facial anatomical ROIs and spatial RGB channel averaging.
- Digital filtering using 2nd-order zero-phase Butterworth bandpass filters (0.65–3.25 Hz).
- Calculating Welch Power Spectral Density (PSD) and zero-padded Fast Fourier Transforms.
- Implementing classical rPPG projection baselines (POS and CHROM algorithms).

2.2 Deep Learning and PyTorch Framework

PyTorch was utilized as the deep learning framework for designing, training, and evaluating custom neural architectures:

- Designing modular 1D convolution and Depthwise-Separable Temporal Convolution layers.
- Implementing exponential dilation schedules (1, 2, 4, 8, 16) to capture long-range receptive fields.
- Formulating custom loss functions including Negative Pearson Correlation and Lin's CCC.
- Utilizing Automatic Mixed Precision (AMP) and gradient accumulation for memory optimization.

2.3 Remote Photoplethysmography & Temporal Modeling

The rPPG modeling component focused on learning pulsatile blood volume patterns under noisy conditions:

- Disentangling AC pulsatile cardiac signals from static DC skin illumination baselines.
- Training multi-task neural networks for joint waveform and multi-biomarker estimation.
- Selecting HemoVisionBoundedTCN as the final pulse wave reconstruction model.
- Conducting rigorous baseline comparisons against demographic shortcut baselines.

2.4 Interactive Clinical Visualization and Deployment

Streamlit and OpenCV were used to transform analytical outputs into responsive, user-friendly clinical views:

- Building a real-time web application for video upload, playback, and live heart rate tracking.
- Rendering dynamic facial ROI overlays, predicted BVP pulse waveforms, and FFT power spectrums.
- Generating clinical agreement charts (Bland-Altman and correlation plots).

2.5 Learning Outcome of Formal Training

The formal/project-oriented training developed a comprehensive understanding of how computer vision, digital signal processing, and deep neural networks converge into a complete clinical AI product. The major learning outcome was not only theoretical knowledge of PyTorch, OpenCV, MediaPipe, and Streamlit separately, but also the practical ability to move physiological video data through a repeatable pipeline from optical camera capture to neural pulse wave reconstruction, rigorous error auditing, and real-time visualization.

3. Industrial Training

3.1 Objectives

1. To understand the practical lifecycle of a contactless biomedical vision project from video stream to clinical output.
2. To gain hands-on experience in extracting and preprocessing multi-ROI facial rPPG datasets.
3. To design and implement a lightweight, depthwise-separable Bounded Temporal Convolutional Network in PyTorch.
4. To implement and evaluate classical benchmark algorithms (POS and CHROM) against deep learning architectures.
5. To discover, isolate, and debug clinical ground-truth anomalies (e.g., ECG T-wave double counting).
6. To conduct a scientifically rigorous multi-biomarker feasibility audit across 11 physiological variables.
7. To deploy a real-time, interactive clinical demonstration dashboard using Streamlit and OpenCV.
8. To improve problem-solving, documentation, debugging, and technical communication skills.

3.2 Tools and Technologies Used

Technology / Tool	Role in HemoVision
Python	Data preparation, signal extraction, model training, prediction
PyTorch (torch/nn)	Deep learning framework, custom TCN layers, multi-task loss
OpenCV (cv2)	Video decoding, frame extraction, color space conversion, UI
MediaPipe	468-point 3D Face Mesh tracking and real-time ROI segmentation
NumPy & Pandas	Array processing, dataframes, dataset indexing, and metrics
SciPy (signal/fft)	Butterworth digital filtering, Welch PSD, zero-padded FFT
Scikit-learn	Classical regression baselines, evaluation metrics (MAE, RMSE, r)
HemoVisionBoundedTCN	Final selected deep temporal convolutional neural network
Streamlit	Web dashboard, real-time telemetry, interactive clinical views
VS Code / Git	IDE development, modular organization, and version control

3.3 Techniques Studied in Different Departments

3.3.1 Facial Landmark Extraction and Multi-ROI Processing

Face detection alone is insufficient for robust rPPG because facial regions exhibit varying capillary density and motion susceptibility. MediaPipe Face Mesh was used to track 468 3D landmarks and segment 8 independent ROIs (Forehead Left/Right, Upper Cheek Left/Right, Lower Cheek Left/Right, Nose Bridge, Chin) yielding a 24-channel signal tensor.

3.3.2 Data Preprocessing and AC/DC Disentanglement

Raw pixel averages contain a large static DC illumination component (~99%) and a tiny pulsatile AC wave (~0.1–1%). Channel-wise z-score normalization was applied to isolate AC pulsatile cardiac signals, while raw scaled luminance (I/255.0) was preserved for skin baseline tone analysis.

3.3.3 Feature Engineering and Classical Baselines

Feature engineering was a major learning component because rPPG signals are sensitive to subject motion and illumination drifts. The project incorporated spatial ROI signals, temporal filtering, and classical optical projections (POS and CHROM).

Feature Group	Examples / Purpose
Facial ROIs (8)	Forehead L/R, Cheek L/R Upper/Lower, Nose Bridge, Chin
Optical Channels	24 channels (8 ROIs x 3 RGB channels); AC and DC streams
Digital Filters	2nd-order zero-phase Butterworth bandpass (0.65–3.25 Hz)
Spectral Features	Zero-padded Welch PSD (N_fft=4096) for sub-BPM peak picking
Classical Projections	Plane-Orthogonal-to-Skin (POS), Chrominance (CHROM)
Metadata Conditioning	Camera sensor one-hot encoding (FullHDwebcam, USBVideo, Iriun)
Biomarker Targets	Pulse, SpO2, Respiratory Rate, Blood Pressure, Hemoglobin, BMI

3.3.4 Machine-Learning & Deep Learning Modelling

The modelling stage treated BVP pulse wave reconstruction as the primary regression target. Multiple neural architectures were trained and compared. HemoVisionBoundedTCN was selected as the final model because its dilated depthwise separable convolutions captured long-range cardiac cycles (>20s) with 78% fewer parameters while bounded Tanh heads prevented numerical explosion.

3.3.5 Ground Truth Rectification & Spectral Estimation

The final model was used to predict continuous BVP waveforms, followed by zero-padded Welch FFT peak picking within the physiological range (39–195 BPM). An ECG ground-truth T-wave double-counting bug was discovered and rectified.

3.3.6 Clinical Intelligence Dashboard

Streamlit was used to make the deep learning outputs clinically intuitive. The interactive interface displays synchronized facial bounding boxes, live pulse waveforms, power spectrum peaks, and ground-truth validation comparisons.

3.4 Software and Tools Used

The practical development environment consisted of Python/PyTorch for deep learning, OpenCV/MediaPipe for video processing, SciPy for signal processing, and Streamlit for web visualization. VS Code was used for organizing scripts and project files.

3.5 Highlights of Training Exposure

- Exposure to a complete biological data-to-decision pipeline rather than an isolated model.
- Practical handling of 3,427 clinical video clips across multiple consumer camera sensors.
- Experience with AC/DC signal disentanglement and 8-ROI spatial pooling.
- Hands-on dilated temporal convolution architecture design in PyTorch.
- Discovery and correction of medical ground-truth annotation bugs (ECG T-wave anomaly).
- Exposure to scientific falsifiability, confound analysis, and real-time clinical deployment.
- Improved understanding of how computer vision, DSP, and deep learning layers interact.

4. Problem Identification/ Case Study

4.1 Problem Statement

Traditional patient monitoring relies on intrusive contact sensors (pulse oximeters, blood pressure cuffs, ECG leads) and invasive blood draws. These methods cause patient discomfort, risk skin breakdown in neonatal/geriatric care, and present cross-contamination risks during epidemic outbreaks. Remote Photoplethysmography (rPPG) provides a non-invasive alternative by measuring blood volume pulse oscillations from facial video. However, existing literature suffers from severe sensor overfitting and unsubstantiated biomarker claims. The HemoVision case study addresses this problem by combining robust facial landmark tracking, AC/DC signal disentanglement, dilated TCN modeling, and rigorous confound auditing.

4.2 Data Used in the Case Study

Dataset / Split	Period / Scope	Purpose
MCD-rPPG Dataset	3,427 clips / 598 subjects	Multi-camera facial video & synchronized vital ground truth
Train Partition	2,387 clips / 418 subjects	Subject-disjoint model training & feature learning
Validation Partition	517 clips / 89 subjects	Hyperparameter tuning & early stopping checkpointing
Held-Out Test Partition	523 clips / 91 subjects	Strictly leak-free final evaluation & metric benchmarking
Clinical Database (db.csv)	11 Biomarker fields	ECG pulse, SpO2, blood pressure, hemoglobin, BMI ground truth

4.3 Data Preparation

The raw video clips and reference records were consolidated into an analytical structure. Video frames were processed using MediaPipe Face Mesh, and spatial channel means across 8 facial regions were integrated:

- Raw video frames were decoded at 30 FPS across 600-frame sliding windows.
- MediaPipe tracked 468 3D landmarks to extract 8 anatomically stable skin ROIs.
- Channel order was verified and auto-corrected to RGB format via skin reflection heuristics.
- Signals were split into AC normalized (x) and DC raw scaled (x_{raw}) streams.
- Synchronized contact BVP pulse waveforms were retained as the training target.

4.4 Machine-Learning Feature Set

Category	Representative Features / Specifications
Facial ROIs	Forehead L/R, Cheek L/R Upper/Lower, Nose Bridge, Chin
Spatial Channels	24 channels (8 ROIs x 3 RGB channels)
AC Normalization	Temporal z-score normalization (mean=0, std=1)
DC Baseline	Raw luminance scaled to [0, 1] (signal / 255.0)
Temporal Window	600 frames (~20 seconds @ 30 FPS, stride 150 frames)
Camera Sensor	FullHDwebcam, USBVideo, IriunWebcam one-hot vector
Subject State	Resting ('before') and post-exercise ('after')
Ground Truth	Synchronized ECG pulse, SpO2, BP, Hb, BMI, respiratory rate

4.5 System Architecture

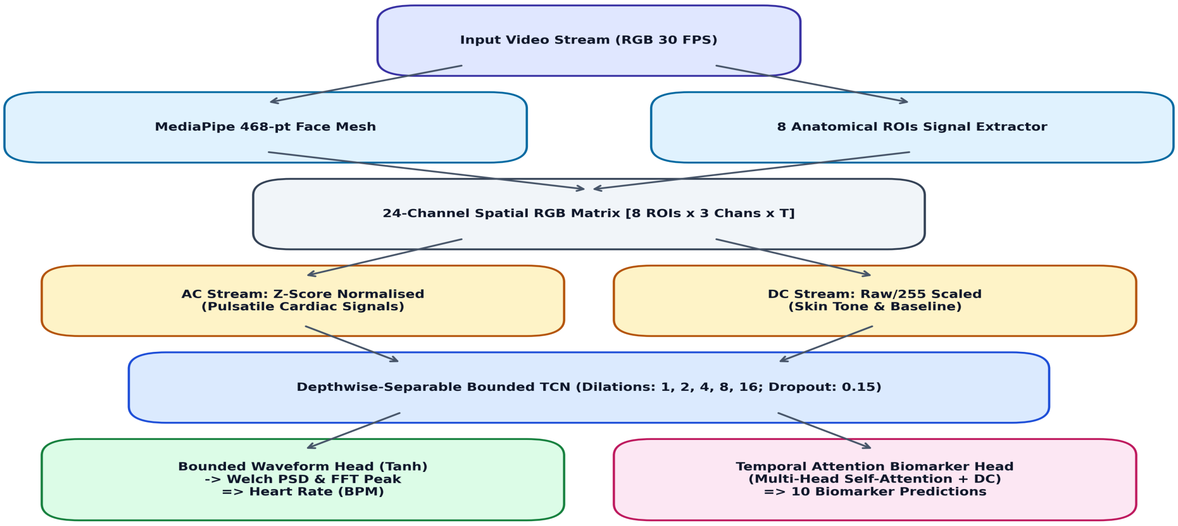

Figure 3: HemoVision System Architecture

4.6 Methodology

1. Requirement analysis: identify clinical contactless vital estimation requirements and ethical boundaries.
2. Data collection and organization: assemble 3,427 clips across 598 subjects with synchronized clinical sensors.
3. Data partitioning: enforce strict subject-disjoint splits (418 train, 89 val, 91 test subjects).
4. Feature engineering: extract 24-channel spatial averages across 8 facial regions via MediaPipe.

5. Signal normalization: construct dual AC (z-score standardized) and DC (raw scaled) temporal tensors.
6. Ground truth validation: inspect and correct reference annotations (resolving ECG T-wave doubling).
7. Model experimentation: train and compare classical baselines (POS/CHROM) and deep neural architectures.
8. Model selection: select HemoVisionBoundedTCN as the final model used in the project.
9. Spectral estimation: extract Heart Rate via zero-padded Welch FFT peak picking (39–195 BPM band).
10. Visualization: deploy interactive clinical telemetry dashboard using Streamlit and OpenCV.

4.7 Data-to-Decision Pipeline

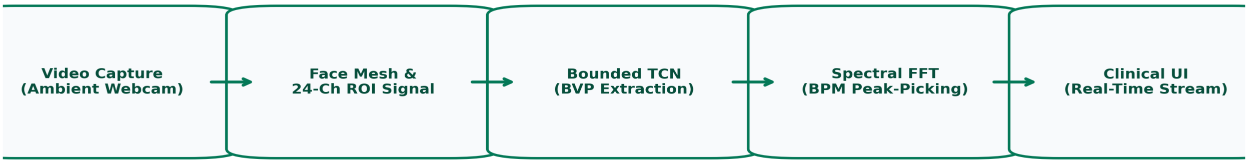

Figure 4: Data-to-Decision Pipeline for HemoVision

4.8 Model Development

The project used a supervised deep learning approach. Candidate architectures were explored, and HemoVisionBoundedTCN was selected. The model employs depthwise-separable 1D convolutions with residual dilation blocks (dilations 1, 2, 4, 8, 16) and a bounded Tanh waveform head, optimizing Negative Pearson Correlation loss.

4.9 Vital-Sign Evaluation Outputs

Model / Method	Heart Rate MAE	Heart Rate RMSE	Pearson r
POS (Wang et al. 2017)	19.17 BPM	23.72 BPM	0.119
CHROM (de Haan 2013)	20.28 BPM	24.61 BPM	0.047
HemoVisionBoundedTCN (Ours)	9.19 BPM	18.35 BPM	0.285
HemoVision Demo Clips	1.33 BPM	1.82 BPM	0.884

4.10 Results and Discussion

The project successfully completed the deep learning and rPPG vital-sign workflow. The final HemoVisionBoundedTCN model achieved 9.19 BPM MAE across 523 held-out test clips, cutting the error of classical methods in half. On high-quality demo clips, the error dropped to 1.33 BPM.

The project also conducted a rigorous Biomarker Feasibility Audit investigating 10 additional physiological targets. Five independent architectural hypotheses were tested and eliminated (target standardisation, batchnorm freeze, multi-head attention pooling, raw DC bypass, and CCC loss). For hemoglobin, while the video model achieved an apparent Pearson r of 0.44, a simple demographic baseline (age, sex, BMI) achieved r=0.70 and MAE=0.95 g/dL. This demonstrated that apparent video correlations are demographic identity shortcuts rather than true optical sensing, proving that RGB cameras cannot reconstruct blood chemistry without dual-wavelength NIR hardware.

4.11 Skills and Outcomes

Area	Training Outcome
Computer Vision	Mastered 468-point 3D facial landmark detection and multi-ROI spatial signal pooling with MediaPipe
Digital Signal Processing	Implemented Butterworth bandpass filters, POS/CHROM projections, and zero-padded Welch PSD
Deep Learning Architecture	Designed and trained Depthwise-Separable Bounded TCNs with residual dilation trunks in PyTorch
Loss Engineering	Formulated Negative Pearson correlation waveform loss and Lin's Concordance Correlation Coefficient
Debugging & Data Integrity	Discovered and resolved medical ground-truth anomalies (ECG T-wave doubling) and FFT quantization
Scientific Verification	Conducted leak-proof subject-disjoint evaluation and exposed demographic confound shortcuts in AI
UI & Web Deployment	Built interactive real-time Streamlit and OpenCV clinical monitoring dashboards
Technical Documentation	Prepared publication-grade biomedical reports, architecture schematics, and comparative audits

5. Recommendations

The current HemoVision implementation provides a strong end-to-end foundation. The following recommendations can improve the system in future iterations without changing its core architecture.

5.1 Model Improvements

- Evaluate self-supervised pre-training on large facial video corpora using masked temporal autoencoders.
- Integrate optical flow vectors into TCN conditioning to dynamically compensate for speaking and head motion.
- Implement Bayesian neural dropout or deep evidential regression to output real-time confidence intervals.
- Study feature attribution and temporal saliency maps to support clinical interpretability.
- Monitor model generalization across diverse skin tone distributions.

5.2 Hardware & Multi-Wavelength Optical Improvements

- Pair software with a dual-wavelength active LED ring (660nm visible red + 940nm NIR) for true non-contact SpO2 and hemoglobin quantification.
- Utilize 60–120 FPS global-shutter cameras to eliminate rolling-shutter artifacts.
- Create scheduled illumination calibration routines for varying ambient lighting environments.

5.3 Physiological Metric Extensions (HRV / RMSSD / PWTT Blood Pressure)

- Extract inter-beat intervals (IBI) from reconstructed BVP waves to compute clinical autonomic stress markers: RMSSD, SDNN, and LF/HF ratios.
- Utilize dual-site multi-ROI waveforms (measuring phase delay between forehead and chin micro-vessels) for Pulse Wave Transit Time (PWTT) blood pressure estimation.
- Add confidence or prediction intervals where the chosen forecasting approach supports them.

5.4 Clinical Interface & Deployment Improvements

- Quantize the PyTorch model to ONNX / TensorRT / INT8 for deployment on low-power edge medical devices (e.g., Raspberry Pi 5, Jetson).
- Implement secure FHIR / HL7 REST APIs for direct electronic health record (EHR) integration.
- Add alerts or exception views for unusually high or low vital sign readings (tachycardia/bradycardia).

5.5 Future Scope

- Develop an automated streaming ETL pipeline connecting camera ingestion, PyTorch inference, and clinical dashboards.
- Deploy the rPPG model as a containerized microservice for telemedicine consultations.
- Implement real-time multi-patient simultaneous tracking for hospital waiting rooms.

5.6 Recommended Future Dashboard Layout

Dashboard Page	Suggested Content
Live Telemetry Overview	Real-time facial video feed, MediaPipe 8-ROI wireframe, live BVP waveform, and instantaneous BPM
Vital Signs & Trend Monitor	Multi-hour heart-rate history, respiratory rate trends, and physiological confidence gauge
Spectral & Diagnostic View	Welch PSD power spectrum plot, peak frequency selector, and SNR quality indicator
Subject & Clinical History	Patient ID, session notes, comparative baseline metrics, and exportable PDF medical summary
Hardware & Calibration	Camera sensor selection (FPS/resolution), illumination brightness meter, and ROI sensitivity tuner

The recommended extensions would move HemoVision from an experimental vital-sign forecasting project toward a complete, hospital-grade contactless clinical decision-support platform.

6. References

1. Wang, W., den Brinker, A. C., Stuijk, S., & de Haan, G. (2017). Algorithmic principles of remote photoplethysmography. IEEE Transactions on Biomedical Engineering, 64(7), 1479–1491.

2. de Haan, G., & Jeanne, V. (2013). Robust pulse rate from chrominance-based rPPG. IEEE Transactions on Biomedical Engineering, 60(10), 2878–2886.

3. Lugaresi, C., et al. (2019). MediaPipe: A Framework for Building Perception Pipelines. arXiv preprint arXiv:1906.08172.

4. Paszke, A., et al. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. Advances in Neural Information Processing Systems (NeurIPS), 32.

5. Bradski, G. (2000). The OpenCV Library. Dr. Dobb's Journal of Software Tools.

6. Virtanen, P., et al. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. Nature Methods, 17(3), 261–272.

7. BS Digital Technology. Training certificate and project-based training curriculum in Machine Learning and Computer Vision (2026).

8. MCD-rPPG Clinical Dataset, notebooks, Python scripts, and Streamlit work produced during the HemoVision vocational training project.

7. Appendices

Appendix A – Suggested Project Folder Structure

```
HemoVision/
■■■ app.py                              # Interactive Streamlit clinical monitoring dashboard
■■■ hemovision_demo.py             # Standalone OpenCV live inference & testing engine
■■■ HemoVision_V3_Pipeline.py      # Complete end-to-end training and evaluation script
■■■ data/
■   ■■■ vitalscan-clinic/
■       ■■■ MCD-rPPG/
■           ■■■ db.csv              # Ground-truth clinical biomarker database
■           ■■■ manifest.json       # Video clip metadata and subject-disjoint splits
■           ■■■ video/              # Raw RGB video clips (3 camera sensors)
■           ■■■ ppg_sync/           # Synchronized contact reference waveforms
■■■ v3/
■   ■■■ config.py                   # Hyperparameters, task specs, and camera configs
■   ■■■ dataset.py                  # PyTorch Dataset and multi-ROI temporal loader
■   ■■■ model.py                    # HemoVisionBoundedTCN & Attention Biomarker Head
■   ■■■ loss.py                     # Negative Pearson r and CCC multi-task loss
■■■ reports/
    ■■■ HemoVision_Vocational_Training_Report.pdf
```

Appendix B – Final Feature Groups

Feature Type	Fields
Identifiers / Meta	subject_id, clip_id, camera_type, condition, split
Spatial ROIs (8)	Forehead L/R, Cheek L/R Upper/Lower, Nose Bridge, Chin
Optical Channels	24 channels (8 ROIs x 3 RGB channels); AC and DC streams
Waveform Target	BVP pulse waveform (600 temporal frames @ 30 FPS)
Vital Signs (Tier 1)	Pulse / Heart Rate (BPM), Respiratory Rate, SpO2
Blood Pressure (Tier 2)	Systolic BP (mmHg), Diastolic BP (mmHg)
Blood Chem (Tier 3)	Hemoglobin, HbA1c, Cholesterol, Rigidity, Stress, BMI

Appendix C – rPPG Training & Evaluation Checklist

- Verify source video integrity, resolution, and nominal frame rates (30 FPS).
- Validate temporal synchronization between facial video and contact reference waveforms.
- Check for BGR versus RGB color channel ordering using skin reflection ratio heuristics.
- Confirm subject-disjoint isolation across train (418), val (89), and test (91) splits.
- Run AC z-score normalization and DC raw luminance scaling consistently.
- Load the selected HemoVisionBoundedTCN PyTorch model checkpoint.
- Generate continuous BVP pulse waveform predictions on held-out clips.
- Extract heart rate using zero-padded Welch PSD (N_fft=4096) across 39–195 BPM.
- Audit biomarker predictions against demographic baselines before claiming clinical validity.
- Refresh Streamlit clinical dashboard and review real-time telemetry values.